Reduced vs Full model, Extra Sum of Squares, General Linear Hypothesis, Partial F test

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Where We Left Off

Tests So Far

- Overall significance: $H_0 : \beta_1 = \cdots = \beta_k = 0$ with $F_0 = \mathrm{MS_R}/\mathrm{MS_{Res}}$.
- Individual coefficient: $H_0 : \beta_j = 0$ with $t_0 = \hat{\beta}_j / \operatorname{se}(\hat{\beta}_j)$.
- Delivery time data, $n = 25$: $\hat{y} = 2.341 + 1.616\, x_1 + 0.0144\, x_2$, $R^2 = 0.9596$.
- Overall test: $F_0 = 261.24$ on $(2, 22)$ degrees of freedom. Distance: $t_0 = 3.98$, $p = 0.0006$.

Beyond One Coefficient at a Time

Questions that a single t-test or the overall F-test cannot answer:

- Does a *group* of regressors improve the fit once the others are already in the model?
- Do two regressors have the same effect, $\beta_j = \beta_k$?
- Is a coefficient, or a combination of coefficients, equal to a specified value?

All three are answered by comparing a *reduced* model with a *full* model.

Roadmap

1. Reduced versus full model.
2. Extra sum of squares.
3. Partial F-test.
4. General linear hypothesis $\mathbf{T}\boldsymbol{\beta} = \mathbf{0}$.
5. Sequential versus partial sums of squares.
6. Added-variable plots.

Running examples: delivery time data (two regressors) and gasoline mileage data (MPV Table B.3, several regressors).

Reduced vs. Full Model

Partitioning the Coefficient Vector

Model with p parameters ($k = p - 1$ regressors):

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Split $\boldsymbol{\beta}$ and the columns of $\mathbf{X}$ into two blocks:

$$\boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 \end{bmatrix}$$

- $\boldsymbol{\beta}_1$ is $(p - r) \times 1$ and contains the intercept.
- $\boldsymbol{\beta}_2$ is $r \times 1$: the r coefficients under test.

The Full Model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

- Uses all p parameters.
- Least-squares fit: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.
- Residual sum of squares $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$ with $n - p$ degrees of freedom.
- Error variance estimate: $\mathrm{MS}_{\mathrm{Res}} = \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n - p)$.

The Hypothesis and the Reduced Model

$$H_0 : \boldsymbol{\beta}_2 = \mathbf{0} \qquad \text{versus} \qquad H_1 : \boldsymbol{\beta}_2 \neq \mathbf{0}$$

If H_0 is true, the regressors in $\mathbf{X}_2$ drop out. The *reduced model* is

$$\mathbf{y} = \mathbf{X}_1 \boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$$

- Least-squares fit: $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$.
- Residual sum of squares $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$ with $n - p + r$ degrees of freedom.

Nested Models

- The reduced model is the full model with $\boldsymbol{\beta}_2$ forced to $\mathbf{0}$.
- Every fit the reduced model can produce is also available to the full model.
- The full model can only match or improve on the reduced model's fit.

$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) \;\geq\; \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$$

The question is whether the improvement is larger than chance alone would produce.

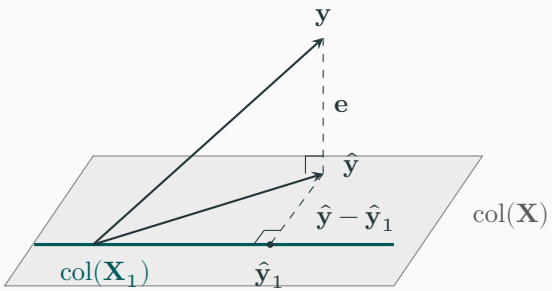
Question: Why Not Just Compare the Residual Sums of Squares?

The full model always has the smaller (or equal) residual sum of squares.

- So can we conclude that $\mathbf{X}_2$ matters whenever $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) < \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$?
- What would a useful comparison need?

Geometry of Nested Models

- $\mathrm{col}(\mathbf{X}_1) \subset \mathrm{col}(\mathbf{X})$.
- $\hat{\mathbf{y}}_1 = \mathbf{H}_1\mathbf{y}$ projects onto the smaller space, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ onto the larger.
- $\hat{\mathbf{y}} - \hat{\mathbf{y}}_1$ is the part of the fit due to $\mathbf{X}_2$.
- $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = \|\mathbf{y} - \hat{\mathbf{y}}_1\|^2$ and $\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \|\mathbf{y} - \hat{\mathbf{y}}\|^2$.



Delivery Time Data: Full vs. Reduced Model

Test whether distance (x_2) adds anything once cases (x_1) are in the model.

Model	Fitted equation	$\mathrm{SS}_{\mathrm{Res}}$	df
Full	$\hat{y} = 2.341 + 1.616\,x_1 + 0.0144\,x_2$	233.73	22
Reduced	$\hat{y} = 3.321 + 2.176\,x_1$	402.13	23

- Here $\boldsymbol{\beta}_1 = (\beta_0, \beta_1)'$ and $\boldsymbol{\beta}_2 = \beta_2$, so $r = 1$.
- $\mathrm{MS}_{\mathrm{Res}}(\mathrm{FM}) = 233.73/22 = 10.62.$

Extra Sum of Squares

Extra Sum of Squares: Definition

Regression sum of squares of the full model, $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$ with $p-1$ df, and of the reduced model, $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_1)$ with $p-r-1$ df. The *extra sum of squares* due to $\boldsymbol{\beta}_2$ given $\boldsymbol{\beta}_1$ is

$$\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) - \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_1) \tag{1}$$

with r degrees of freedom.

Interpreting the Extra Sum of Squares

- It is the increase in the regression sum of squares when $\mathbf{X}_2$ is added to a model that already contains $\mathbf{X}_1$.
- The word “given” matters: the contribution of $\mathbf{X}_2$ depends on what is already in the model.
- Equivalently, it is how much of the unexplained variation $\mathbf{X}_2$ removes.

Extra Sum of Squares as a Reduction in $\mathrm{SS}_{\mathrm{Res}}$

Both models decompose the same total sum of squares:

$$\mathrm{SS}_{\mathrm{T}} = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) + \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})$$
$$\mathrm{SS}_{\mathrm{T}} = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_1) + \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$$

Subtracting the second line from the first and using (1):

$$\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1) = \mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) \tag{2}$$

Partitioning the Regression Sum of Squares

Rearranging (1):

$$\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_1) + \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)$$

- The regression sum of squares splits into what $\mathbf{X}_1$ explains and what $\mathbf{X}_2$ adds.
- Degrees of freedom split the same way: $p - 1 = (p - r - 1) + r$.

Delivery Time Data: Extra Sum of Squares for Distance

$$\begin{aligned}\mathrm{SS_R}(\beta_2 \mid \beta_1, \beta_0) &= \mathrm{SS_R}(\boldsymbol{\beta}) - \mathrm{SS_R}(\boldsymbol{\beta}_1) = 5550.81 - 5382.41 = 168.40 \\ &= \mathrm{SS_{Res}}(\mathrm{RM}) - \mathrm{SS_{Res}}(\mathrm{FM}) = 402.13 - 233.73 = 168.40\end{aligned}$$

- Both routes agree, as (2) requires.
- Adding distance to a model with cases reduces the residual sum of squares by 168.40, from 402.13 to 233.73.

Distribution Under H_0

Assume $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$. If $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$ is true:

- $\mathrm{SS_R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)/\sigma^2 \sim \chi^2_r$.
- $\mathrm{SS_{Res}}(\mathrm{FM})/\sigma^2 \sim \chi^2_{n-p}$.
- The two are independent.

The next frames show why, using the hat matrices of the two models.

Projection Form of the Extra Sum of Squares

Hat matrices of the full and reduced models:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \mathbf{H}_1 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'$$

Residual sums of squares are quadratic forms in $\mathbf{y}$:

$$\begin{aligned}
\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) &= \mathbf{y}'(\mathbf{I} - \mathbf{H})\mathbf{y} \\
\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) &= \mathbf{y}'(\mathbf{I} - \mathbf{H}_1)\mathbf{y} \\
\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1) &= \mathbf{y}'(\mathbf{H} - \mathbf{H}_1)\mathbf{y}
\end{aligned}$$

Why $\mathbf{H} - \mathbf{H}_1$ Is a Projection: Step 1

$\text{col}(\mathbf{X}_1) \subset \text{col}(\mathbf{X})$, so projecting a vector already in $\text{col}(\mathbf{X}_1)$ onto $\text{col}(\mathbf{X})$ leaves it unchanged: $\mathbf{H}\mathbf{X}_1 = \mathbf{X}_1$. Hence

$$\mathbf{H}\mathbf{H}_1 = \mathbf{H}\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1' = \mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1' = \mathbf{H}_1$$
$$\mathbf{H}_1\mathbf{H} = (\mathbf{H}\mathbf{H}_1)' = \mathbf{H}_1$$

Why $\mathbf{H} - \mathbf{H}_1$ Is a Projection: Step 2

$$(\mathbf{H} - \mathbf{H}_1)^2 = \mathbf{H} - \mathbf{H}\mathbf{H}_1 - \mathbf{H}_1\mathbf{H} + \mathbf{H}_1 = \mathbf{H} - \mathbf{H}_1$$
$$\operatorname{rank}(\mathbf{H} - \mathbf{H}_1) = \operatorname{tr}(\mathbf{H} - \mathbf{H}_1) = p - (p - r) = r$$

- $\mathbf{H} - \mathbf{H}_1$ is symmetric and idempotent of rank r: a projection onto an r-dimensional space.
- Also $(\mathbf{H} - \mathbf{H}_1)(\mathbf{I} - \mathbf{H}) = \mathbf{0}$, so the two quadratic forms involve orthogonal pieces of $\mathbf{y}$ and are independent.
- Under H_0, $\mathrm{E}[\mathbf{y}] = \mathbf{X}_1\boldsymbol{\beta}_1$ is annihilated (sent to $\mathbf{0}$) by $\mathbf{H} - \mathbf{H}_1$, so the form depends on $\boldsymbol{\varepsilon}$ only and is $\sigma^2\chi^2_r$.

Background: What Does “Annihilate” Mean?

A matrix $\mathbf{A}$ *annihilates* a vector $\mathbf{v}$ if

$$\mathbf{A}\mathbf{v} = \mathbf{0}$$

- Nothing of $\mathbf{v}$ survives the multiplication.
- The set of all vectors that $\mathbf{A}$ annihilates is its *null space*.
- Familiar case: $\mathbf{I} - \mathbf{H}$ annihilates every vector in $\mathrm{col}(\mathbf{X})$, since $(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$. This is why the residuals $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ contain no trace of $\mathrm{E}[\mathbf{y}]$.
- Here: $\mathbf{H}\mathbf{X}_1 = \mathbf{X}_1 = \mathbf{H}_1\mathbf{X}_1$, so $\mathbf{H} - \mathbf{H}_1$ annihilates $\mathrm{col}(\mathbf{X}_1)$, which contains $\mathrm{E}[\mathbf{y}] = \mathbf{X}_1\boldsymbol{\beta}_1$ under H_0.

Question: Why Does the Extra Sum of Squares Have r Degrees of Freedom?

- We added r columns to the model. How does that show up in the projection picture?

Partial F-Test

The Partial F Statistic

$$F_0 = \frac{\mathrm{SS_R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)/r}{\mathrm{MS_{Res}}} \tag{3}$$

- $\mathrm{MS_{Res}}$ comes from the *full* model.
- Under $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$, $F_0 \sim F_{r,\, n-p}$.
- Reject H_0 at level α if $F_0 > F_{\alpha,\, r,\, n-p}$.

Partial F Statistic from the Residual Sums of Squares

Substituting (2) into (3):

$$F_0 = \frac{\left[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})\right]/r}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \tag{4}$$

Only the two residual sums of squares and their degrees of freedom are needed.

Reading the Statistic

- Numerator: the average reduction in $\mathrm{SS}_{\mathrm{Res}}$ per added parameter.
- Denominator: the noise level σ^2, estimated from the full model.
- $F_0 \approx 1$: the added regressors reduce $\mathrm{SS}_{\mathrm{Res}}$ no more than noise would.
- F_0 large: the reduction is too big to be chance, so reject H_0.

ANOVA Table with Extra-Sum-of-Squares Rows

Source	SS	df	MS	F_0
Regression	$\mathrm{SS_R}(\boldsymbol{\beta})$	$p-1$	$\mathrm{MS_R}$	$\mathrm{MS_R}/\mathrm{MS_{Res}}$
$\boldsymbol{\beta}_1$	$\mathrm{SS_R}(\boldsymbol{\beta}_1)$	$p-r-1$		
$\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1$	$\mathrm{SS_R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)$	r	$\mathrm{SS_R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)/r$	$\dfrac{\mathrm{SS_R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)/r}{\mathrm{MS_{Res}}}$
Residual	$\mathrm{SS_{Res}}$	$n-p$	$\mathrm{MS_{Res}}$	
Total	$\mathrm{SS_T}$	$n-1$		

Special Case: The Overall F-Test

Take $\boldsymbol{\beta}_1 = \beta_0$ and $\boldsymbol{\beta}_2 = (\beta_1, \ldots, \beta_k)'$, so $r = p - 1 = k$.

- Reduced model: intercept only, $\hat{y}_i = \bar{y}$, so $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = \mathrm{SS}_{\mathrm{T}}$.
- Extra sum of squares: $\mathrm{SS}_{\mathrm{T}} - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$.

$$F_0 = \frac{\mathrm{SS}_{\mathrm{R}}/k}{\mathrm{MS}_{\mathrm{Res}}} = \frac{\mathrm{MS}_{\mathrm{R}}}{\mathrm{MS}_{\mathrm{Res}}}$$

The overall test is the partial F-test with every slope in the tested block.

Special Case: One Coefficient

Take $r = 1$, so $\boldsymbol{\beta}_2 = \beta_j$. With $C_{jj} = [(\mathbf{X}'\mathbf{X})^{-1}]_{jj}$:

$$\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{all other } \beta\text{'s}) = \frac{\hat{\beta}_j^2}{C_{jj}}$$

$$F_0 = \frac{\hat{\beta}_j^2 / C_{jj}}{\mathrm{MS}_{\mathrm{Res}}} = \left(\frac{\hat{\beta}_j}{\sqrt{\mathrm{MS}_{\mathrm{Res}}\, C_{jj}}} \right)^2 = t_0^2$$

The partial F-test for a single coefficient is the two-sided t-test, since $F_{1,\nu} = t_\nu^2$.

Delivery Time Data: Does Distance Add to Cases?

$H_0 : \beta_2 = 0$ versus $H_1 : \beta_2 \neq 0$, given x_1 in the model.

	$\mathrm{SS}_{\mathrm{Res}}$	df
Full: x_1, x_2	233.73	22
Reduced: x_1	402.13	23
Difference	168.40	1

Delivery Time Data: Partial F for Distance

$$F_0 = \frac{168.40/1}{233.73/22} = \frac{168.40}{10.62} = 15.85$$

- $F_{0.05,1,22} = 4.30$ and $p = 0.0006$: reject H_0.
- Distance contributes significantly given cases.
- Check: $t_0 = 3.98$ for β_2 and $t_0^2 = 15.85 = F_0$.

Delivery Time Data: Does Cases Add to Distance?

Reverse the roles: $H_0 : \beta_1 = 0$, with x_2 in the reduced model.

$$F_0 = \frac{(1185.39 - 233.73)/1}{233.73/22} = \frac{951.66}{10.62} = 89.58$$

- $p = 3.3 \times 10^{-9}$: reject H_0.
- Check: $t_0 = 9.46$ for β_1 and $t_0^2 = 89.58$.

A Second Dataset: Gasoline Mileage

MPV Table B.3: 32 automobiles, response y = miles per gallon.

- Regressors used here: horsepower, torque, rear axle ratio, carburetor barrels, weight.
- Torque is missing for two cars, so we use the $n = 30$ complete cases.

$$y = \beta_0 + \beta_{\mathrm{hp}} x_{\mathrm{hp}} + \beta_{\mathrm{tq}} x_{\mathrm{tq}} + \beta_{\mathrm{ax}} x_{\mathrm{ax}} + \beta_{\mathrm{cb}} x_{\mathrm{cb}} + \beta_{\mathrm{wt}} x_{\mathrm{wt}} + \varepsilon$$

Here $p = 6$, so the residual degrees of freedom are $n - p = 24$.

Gasoline Mileage: Full Model Fit

Term	Estimate	Std. error	t_0	p-value
Intercept	29.4244	7.0840	4.154	0.0004
Horsepower	-0.0204	0.0714	-0.285	0.778
Torque	-0.0240	0.0525	-0.457	0.652
Rear axle ratio	1.9610	2.0862	0.940	0.357
Carburetor barrels	0.4261	0.9560	0.446	0.660
Weight	-0.00235	0.00192	-1.224	0.233

$\mathrm{SS}_{\mathrm{R}} = 860.48$ (5 df), $\mathrm{SS}_{\mathrm{Res}} = 278.62$ (24 df), $\mathrm{MS}_{\mathrm{Res}} = 11.609$, $R^2 = 0.755$, overall $F_0 = 14.82$ ($p \approx 10^{-6}$).

Question: Every t-Test Is Insignificant, Yet the Overall F Is Large

- How can the model be highly significant while no individual coefficient is?
- Sample correlations: horsepower–torque 0.96, torque–weight 0.91, horsepower–weight 0.85.

Gasoline Mileage: Testing a Group of Regressors

$H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \beta_{\text{wt}} = 0$, so $r = 3$: do the three engine-size and weight measures add anything beyond axle ratio and carburetor?

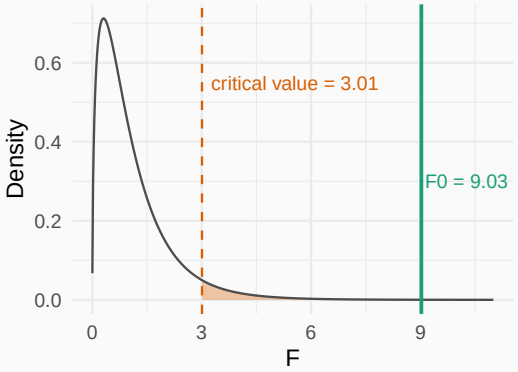
	SS_{Res}	df
Full: hp, tq, ax, cb, wt	278.62	24
Reduced: ax, cb	592.96	27
Difference	314.34	3

Reduced fit: $\widehat{y} = 3.768 + 6.718\, x_{\text{ax}} - 1.713\, x_{\text{cb}}$.

Gasoline Mileage: Partial F Calculation

$$F_0 = \frac{314.34/3}{278.62/24} = \frac{104.78}{11.609} = 9.03$$

- $F_{0.05,3,24} = 3.01$.
- $p = 0.0004$: reject H_0.
- The group matters even though each member has $p > 0.2$ individually.



Gasoline Mileage: A Group That Does Not Help

$H_0 : \beta_{\text{ax}} = \beta_{\text{cb}} = 0$, so $r = 2$, with hp, tq, wt kept in the model.

$$F_0 = \frac{15.03/2}{278.62/24} = \frac{7.514}{11.609} = 0.65$$

- $F_{0.05,2,24} = 3.40$ and $p = 0.53$: do not reject H_0.
- Given horsepower, torque and weight, axle ratio and carburetor add nothing detectable.

Question: Why One Partial F Instead of Several t-Tests?

- For $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \beta_{\text{wt}} = 0$ we could run three t-tests. Why is that a poor substitute for $F_0 = 9.03$?

General Linear Hypothesis

Linear Restrictions on the Coefficients

$$H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{0} \qquad \text{versus} \qquad H_1 : \mathbf{T}\boldsymbol{\beta} \neq \mathbf{0}$$

- $\mathbf{T}$ is an $r \times p$ matrix of rank r.
- Each row of $\mathbf{T}$ is one linear restriction on $\boldsymbol{\beta}$.
- Rank r means the r restrictions are independent: none is implied by the others.

Examples of $\mathbf{T}$ for the Gasoline Model

Parameter order $(\beta_0, \beta_{\text{hp}}, \beta_{\text{tq}}, \beta_{\text{ax}}, \beta_{\text{cb}}, \beta_{\text{wt}})'$.

Hypothesis	r	$\mathbf{T}$
$\beta_{\text{wt}} = 0$	1	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$
$\beta_{\text{hp}} = \beta_{\text{tq}} = 0$	2	$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$
$\beta_{\text{hp}} = \beta_{\text{tq}}$	1	$\begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \end{bmatrix}$
$\beta_{\text{hp}} = \beta_{\text{tq}},\ \beta_{\text{ax}} = 0$	2	$\begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$

Subset Tests Are a Special Case

If the tested block $\boldsymbol{\beta}_2$ is placed last:

$$\mathbf{T} = \begin{bmatrix} \mathbf{0}_{r\times(p-r)} & \mathbf{I}_r \end{bmatrix} \quad \Longrightarrow \quad \mathbf{T}\boldsymbol{\beta} = \boldsymbol{\beta}_2$$

- $H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{0}$ is exactly $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$.
- Single-coefficient, subset and overall tests are all general linear hypotheses.
- The general form also covers restrictions that are not of the form “coefficients equal zero”.

Obtaining the Reduced Model

Impose the restrictions in the full model and simplify.

Example: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}} = \gamma$. Substituting,

$$y = \beta_0 + \gamma\,(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{ax}} x_{\text{ax}} + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}} + \varepsilon$$

- The two columns are replaced by their sum.
- The reduced model has $p - r = 5$ parameters and is fit by ordinary least squares.

Test Statistic for the General Linear Hypothesis

$$F_0 = \frac{[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})]/r}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \tag{5}$$

- Same form as (4); $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM})$ is now the residual sum of squares under the restrictions.
- Under H_0, $F_0 \sim F_{r,\,n-p}$. Reject if $F_0 > F_{\alpha,\,r,\,n-p}$.

Matrix Form of the Test

The difference in residual sums of squares can be computed from the full-model fit alone:

$$\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM}) = \mathrm{SS}_{\mathrm{H}} = (\mathbf{T}\hat{\boldsymbol{\beta}})' \left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1} (\mathbf{T}\hat{\boldsymbol{\beta}}) \qquad (6)$$

- $F_0 = (\mathrm{SS}_{\mathrm{H}}/r)/\mathrm{MS}_{\mathrm{Res}}$, the same statistic as (5).
- No reduced model needs to be fit.

Why the Quadratic Form Works

- $\mathrm{Var}(\mathbf{T}\hat{\boldsymbol{\beta}}) = \sigma^2\,\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'$.
- Under H_0, $\mathbf{T}\hat{\boldsymbol{\beta}}$ has mean $\mathbf{0}$, so $\mathrm{SS}_{\mathrm{H}}/\sigma^2$ is a standardized squared length of an r-dimensional normal vector: χ^2_r.
- For $r = 1$, $\mathrm{SS}_{\mathrm{H}}/\mathrm{MS}_{\mathrm{Res}}$ reduces to $(\mathbf{T}\hat{\boldsymbol{\beta}})^2/(\mathrm{MS}_{\mathrm{Res}}\,\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}') = t_0^2$.

Gasoline Mileage: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ by the Matrix Form

$\mathbf{T} = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 \end{bmatrix}$, $r = 1$.

$$\mathbf{T}\hat{\boldsymbol{\beta}} = \hat{\beta}_{\text{hp}} - \hat{\beta}_{\text{tq}} = -0.02039 - (-0.02397) = 0.003582$$

$$\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}' = 0.001217$$

$$\text{SS}_{\text{H}} = \frac{0.003582^2}{0.001217} = 0.0105$$

$$F_0 = \frac{0.0105/1}{11.609} = 0.0009$$

$F_{0.05,1,24} = 4.26$ and $p = 0.976$: do not reject.

Gasoline Mileage: $H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ by the Reduced Model

Fit $y = \beta_0 + \gamma(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{ax}} x_{\text{ax}} + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}}$:

$$\begin{aligned} \hat{y} &= 29.352 - 0.02248\,(x_{\text{hp}} + x_{\text{tq}}) \\ &\quad + 1.999\,x_{\text{ax}} + 0.4307\,x_{\text{cb}} - 0.002374\,x_{\text{wt}} \\ \text{SS}_{\text{Res}}(\text{RM}) &= 278.634 \quad (25\ \text{df}) \end{aligned}$$

- $\text{SS}_{\text{Res}}(\text{RM}) - \text{SS}_{\text{Res}}(\text{FM}) = 278.634 - 278.623 = 0.0105$, matching SS_{H}.
- The data are consistent with horsepower and torque having the same coefficient.

Gasoline Mileage: Two Restrictions at Once

$H_0 : \beta_{\text{hp}} = \beta_{\text{tq}}$ and $\beta_{\text{ax}} = 0$, so $r = 2$. The reduced model has three predictor terms:

$$y = \beta_0 + \gamma\,(x_{\text{hp}} + x_{\text{tq}}) + \beta_{\text{cb}} x_{\text{cb}} + \beta_{\text{wt}} x_{\text{wt}} + \varepsilon$$

$$\text{SS}_{\text{Res}}(\text{RM}) = 295.24 \ (26 \text{ df})$$

$$F_0 = \frac{(295.24 - 278.62)/2}{11.609} = \frac{8.308}{11.609} = 0.72$$

$F_{0.05,2,24} = 3.40$ and $p = 0.50$: do not reject.

Nonzero Right-Hand Side: $\mathbf{T}\boldsymbol{\beta} = \mathbf{c}$

$$H_0 : \mathbf{T}\boldsymbol{\beta} = \mathbf{c} \qquad \Longrightarrow \qquad \mathrm{SS}_{\mathrm{H}} = (\mathbf{T}\hat{\boldsymbol{\beta}} - \mathbf{c})' \left[\mathbf{T}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{T}'\right]^{-1} (\mathbf{T}\hat{\boldsymbol{\beta}} - \mathbf{c})$$

Reduced model: move the known part to the left-hand side.

- Example: $H_0 : \beta_{\mathrm{cb}} = 1$. Regress $y - x_{\mathrm{cb}}$ on the other regressors: $\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) = 282.81$.
- $F_0 = (282.81 - 278.62)/11.609 = 0.36$ and $p = 0.55$.
- Check: $t_0 = (0.4261 - 1)/0.9560 = -0.600$ and $t_0^2 = 0.36$.

Summary of Special Cases

Hypothesis	$\mathbf{T}$ rows	r	Reduced model
Single coefficient	one unit row	1	Drop the regressor
Subset	r unit rows	r	Drop the subset
All slopes	k unit rows	k	Intercept only
Equal coefficients	$(\dots, 1, \dots, -1, \dots)$	1	Sum the two columns
Several restrictions	r independent rows	r	Substitute all restrictions

All use (5) with $F_{r,\, n-p}$.

Question: Is $\beta_{\text{hp}} = \beta_{\text{tq}}$ Meaningful?

- Horsepower and torque are on different scales. Is a test of equal coefficients meaningful?
- How could the hypothesis be made scale-free?

Sequential vs. Partial Sums of Squares

Sequential Sums of Squares

Enter the regressors in a chosen order $x_1, x_2, \dots, x_k$. The sequential sums of squares are

$$\mathrm{SS_R}(\beta_1 \mid \beta_0), \quad \mathrm{SS_R}(\beta_2 \mid \beta_0, \beta_1), \quad \dots, \quad \mathrm{SS_R}(\beta_k \mid \beta_0, \dots, \beta_{k-1})$$

- Each is an extra sum of squares with $r = 1$.
- Each regressor is adjusted only for those *entered before it*.
- Software often calls these Type I sums of squares.

Sequential Sums of Squares Add Up

Applying (1) one regressor at a time telescopes:

$$\mathrm{SS_R}(\boldsymbol{\beta}) = \sum_{j=1}^{k} \mathrm{SS_R}(\beta_j \mid \beta_0, \ldots, \beta_{j-1}) \tag{7}$$

- The decomposition accounts for the whole regression sum of squares.
- It depends on the order of entry.

Partial Sums of Squares

$$\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \beta_0, \dots, \beta_{j-1}, \beta_{j+1}, \dots, \beta_k)$$

- Each regressor is adjusted for *all* the others, as if it entered last.
- Equal to $t_0^2 \, \mathrm{MS}_{\mathrm{Res}}$ for the individual t-test, hence the partial F-test with $r = 1$.
- Software often calls these Type III sums of squares.
- In general they do not add up to $\mathrm{SS}_{\mathrm{R}}(\boldsymbol{\beta})$.

Delivery Time Data: Sequential vs. Partial

	Sequential		Partial
	x_1 first	x_2 first	
Cases (x_1)	5382.41	951.66	951.66
Distance (x_2)	168.40	4599.15	168.40
Sum	5550.81	5550.81	1120.06

- Both sequential columns add to $\mathrm{SS_R}(\boldsymbol{\beta}) = 5550.81$; the partial column does not.
- The size of each regressor's share depends on the order.

Question: Which Sequential Sum of Squares Equals the Partial One?

- In the delivery table, which sequential entries coincide with the partial sums of squares?
- Why?

Gasoline Mileage: Sequential vs. Partial

	Sequential		Partial
	hp, tq, ax, cb, wt	wt, cb, ax, tq, hp	
Horsepower	723.26	0.95	0.95
Torque	107.17	18.71	2.42
Rear axle ratio	2.78	56.81	10.26
Carburetor	9.87	4.71	2.31
Weight	17.39	779.31	17.39
Sum	860.48	860.48	33.32

Horsepower carries 723 when entered first and 0.95 when entered last: the regressors overlap heavily.

When Sequential Equals Partial: Orthogonal Columns

If $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$, then $\mathrm{SS}_\mathrm{R}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1) = \mathrm{SS}_\mathrm{R}(\boldsymbol{\beta}_2)$: the contribution of each block is unaffected by the other, in any order.

- Delivery data: replace x_2 by $z = e(x_2 \mid x_1)$, the part of distance not explained by cases (orthogonal to $\mathbf{1}$ and x_1).
- Regress y on x_1 and z: sequential sums of squares are 5382.41 for x_1 and 168.40 for z in either order.
- 168.40 is exactly $\mathrm{SS}_\mathrm{R}(\beta_2 \mid \beta_1, \beta_0)$ from the original fit.

The same z will reappear as the horizontal axis of an added-variable plot.

Software Output

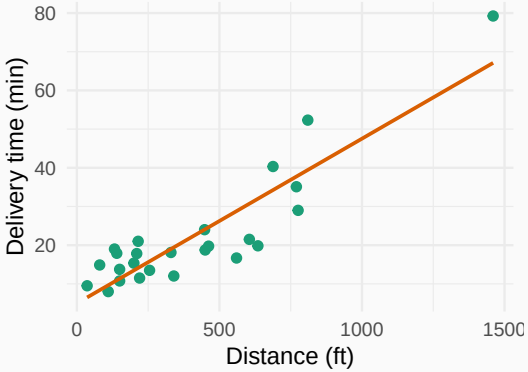
	Sequential	Partial
R	`anova(fit)`	`drop1(fit, test = "F")` `car::Anova(fit, type = "III")`

- `anova()` depends on the order of terms in the model formula.
- The F-statistics in the partial output equal t_0^2 for each coefficient.
- Check which type your software reports by default before interpreting a table.

Added-Variable Plots

Why Look at a Plot?

- A scatterplot of y against x_j ignores the other regressors.
- Delivery data: the marginal slope of time on distance is 0.0426 min/ft.
- The coefficient of distance in the full model is 0.0144 min/ft.
- We want a plot showing the effect of x_j *after* adjusting for the rest.



Constructing an Added-Variable Plot

Let $\mathbf{X}_{(j)}$ be the design matrix with the column of x_j removed (intercept kept).

1. Regress y on $\mathbf{X}_{(j)}$ and keep the residuals $e(y \mid \mathbf{X}_{(j)})$.
2. Regress x_j on $\mathbf{X}_{(j)}$ and keep the residuals $e(x_j \mid \mathbf{X}_{(j)})$.
3. Plot $e(y \mid \mathbf{X}_{(j)})$ against $e(x_j \mid \mathbf{X}_{(j)})$.

Also called a partial regression plot.

Residuals as Adjusted Variables

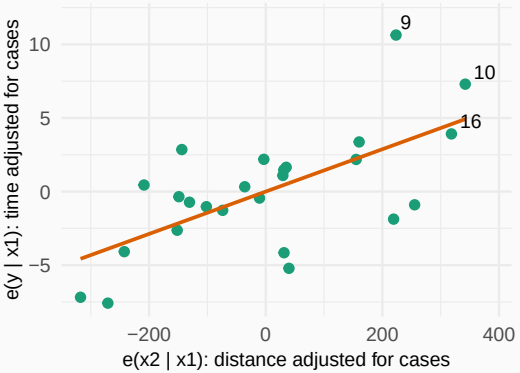
With $\mathbf{H}_{(j)} = \mathbf{X}_{(j)}(\mathbf{X}'_{(j)}\mathbf{X}_{(j)})^{-1}\mathbf{X}'_{(j)}$:

$$e(y \mid \mathbf{X}_{(j)}) = (\mathbf{I} - \mathbf{H}_{(j)})\mathbf{y}$$
$$e(x_j \mid \mathbf{X}_{(j)}) = (\mathbf{I} - \mathbf{H}_{(j)})\mathbf{x}_j$$

- The first is the part of y not explained by the other regressors.
- The second is the part of x_j not explained by the other regressors.

Delivery Time Data: Added-Variable Plot for Distance

- Adjusted for cases, longer distances go with longer delivery times.
- The fitted line has slope 0.01438, the coefficient $\hat{\beta}_2$ from the full model.
- Labelled points: 9 has the largest residual; 10 and 16 have the most extreme adjusted distance.



Key Property: The Slope Is $\widehat{\beta}_j$

The least-squares line through the added-variable plot has:

- Slope $\widehat{\beta}_j$, the coefficient from the full model.
- Intercept 0.
- Residuals equal to the full-model residuals $\mathbf{e}$.

The next two frames show why.

Derivation, Step 1: Decompose y

The full-model fit separates x_j from the other regressors:

$$\mathbf{y} = \mathbf{X}_{(j)}\hat{\boldsymbol{\beta}}_{(j)} + \mathbf{x}_j\hat{\beta}_j + \mathbf{e}$$

Apply $\mathbf{I} - \mathbf{H}_{(j)}$. It annihilates $\mathbf{X}_{(j)}$, and $(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{e} = \mathbf{e}$ because $\mathbf{e} \perp \mathrm{col}(\mathbf{X}) \supset \mathrm{col}(\mathbf{X}_{(j)})$:

$$e(y \mid \mathbf{X}_{(j)}) = \hat{\beta}_j\, e(x_j \mid \mathbf{X}_{(j)}) + \mathbf{e} \tag{8}$$

Derivation, Step 2: Slope and Residuals

Both adjusted variables have mean zero, so the line through the plot has no intercept. Its slope is

$$\frac{e(x_j \mid \mathbf{X}_{(j)})' \, e(y \mid \mathbf{X}_{(j)})}{e(x_j \mid \mathbf{X}_{(j)})' \, e(x_j \mid \mathbf{X}_{(j)})} = \hat{\beta}_j + \frac{e(x_j \mid \mathbf{X}_{(j)})' \, \mathbf{e}}{e(x_j \mid \mathbf{X}_{(j)})' \, e(x_j \mid \mathbf{X}_{(j)})}$$

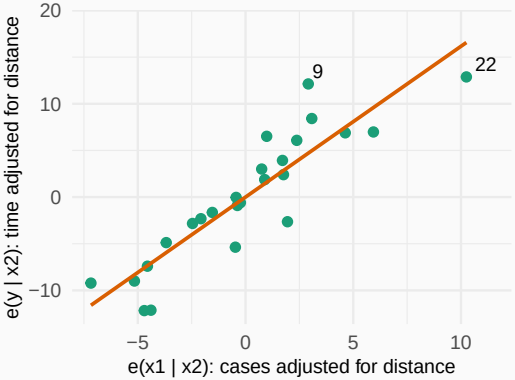
The numerator of the last term is $\mathbf{x}_j'(\mathbf{I} - \mathbf{H}_{(j)})\mathbf{e} = \mathbf{x}_j'\mathbf{e} = 0$. So the slope is $\hat{\beta}_j$ and, by (8), the residuals about the line are $\mathbf{e}$.

Consequences

- The plot has the same residuals, hence the same residual sum of squares, as the full model.
- The sum of squares explained by the line is $\hat{\beta}_j^2 \sum_i e_i(x_j \mid \mathbf{X}_{(j)})^2 = \mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{others})$.
- Delivery data, distance: $0.014385^2 \times 813{,}838.7 = 168.40$.
- Outliers and influential points for $\hat{\beta}_j$ can be seen directly in the plot.

Delivery Time Data: Added-Variable Plot for Cases

- Slope 1.616, the full-model coefficient $\hat{\beta}_1$.
- The sum of squares due to the line is $951.66 = \mathrm{SS}_{\mathrm{R}}(\beta_1 \mid \beta_2, \beta_0)$.
- Point 22 has the most extreme adjusted number of cases.
- Point 9 again has the largest residual.



Squared Correlation in the Plot

The squared correlation between the two adjusted variables is

$$r^2 = \frac{\hat{\beta}_j^2 \sum e(x_j \mid \mathbf{X}_{(j)})^2}{\sum e(y \mid \mathbf{X}_{(j)})^2} = \frac{\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{others})}{\mathrm{SS}_{\mathrm{Res}}(\text{model without } x_j)}$$

- Distance: $168.40/402.13 = 0.419$, correlation 0.647.
- Cases: $951.66/1185.39 = 0.803$, correlation 0.896.
- This is the partial correlation of y and x_j given the other regressors; it returns in model selection.

Collinearity in the Plot

The horizontal spread of the plot is what is left of x_j after the other regressors take their share:

$$\sum_i e_i(x_j \mid \mathbf{X}_{(j)})^2 = (1 - R_j^2) \sum_i (x_{ij} - \bar{x}_j)^2, \qquad \operatorname{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\sum_i e_i(x_j \mid \mathbf{X}_{(j)})^2}$$

- Delivery data: the variance of distance is $105{,}747$, but only $33{,}910$ remains after adjusting for cases, a fraction $0.321 = 1 - 0.824^2$.
- Strong collinearity gives a narrow plot and an imprecise slope.

What to Look for in an Added-Variable Plot

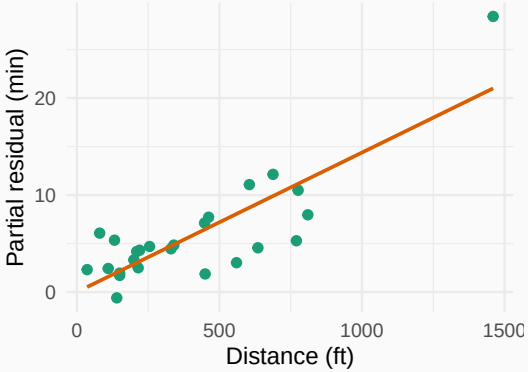
- Linearity: a straight band supports a linear term in x_j.
- Points far from the line: outliers after adjusting for the other regressors.
- Points extreme horizontally: potentially influential for $\hat{\beta}_j$.
- A narrow horizontal spread: collinearity between x_j and the other regressors.

Question: A Point Far from the Line

- A point lies far from the line in the added-variable plot but looks ordinary in the raw scatterplot of y against x_j.
- What does this tell you?

Partial Residual Plot: A Companion

- Plot $e_i + \hat{\beta}_j x_{ij}$ against x_{ij}.
- Slope is $\hat{\beta}_j$, as in the added-variable plot.
- The horizontal axis keeps the original scale of x_j.
- Curvature in the effect of x_j is easier to see here.



Limits of Added-Variable Plots

- They show the partial effect of one regressor under a linear model for the others.
- They are not the best tool for deciding which transformation of x_j is needed; use a partial residual plot for that.
- A relationship between x_j and the other regressors that is not linear can distort the picture.
- They are a diagnostic, not a test: the F-test of β_j remains the formal tool.

Summary

Summary of Tests

Test	Hypothesis	r	Reduced model
Overall	$\beta_1 = \cdots = \beta_k = 0$	k	Intercept only
Individual	$\beta_j = 0$	1	Drop x_j
Subset (partial F)	$\boldsymbol{\beta}_2 = \mathbf{0}$	r	Drop $\mathbf{X}_2$
General linear	$\mathbf{T}\boldsymbol{\beta} = \mathbf{0}$	r	Impose restrictions

All use $F_0 = \dfrac{[\mathrm{SS}_{\mathrm{Res}}(\mathrm{RM}) - \mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})]/r}{\mathrm{SS}_{\mathrm{Res}}(\mathrm{FM})/(n-p)} \sim F_{r,\,n-p}.$

Key Takeaways

- A full-versus-reduced comparison tests any group of coefficients through one F statistic.
- The extra sum of squares is the drop in $\mathrm{SS}_{\mathrm{Res}}$, with one degree of freedom per added parameter.
- Individual t-tests can all be insignificant while a group is highly significant.
- Sequential sums of squares depend on order and add up; partial sums of squares adjust for everything else and do not.
- An added-variable plot shows $\hat{\beta}_j$, the full-model residuals and $\mathrm{SS}_{\mathrm{R}}(\beta_j \mid \text{others})$ in one picture.

Questions?